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Cor Caratheodory Coeficientes Taylor Fn Holomorfa Parte Real Positiva

Corolario 1 (Carathéodory). *Sea $f \in \mathcal{H}(\mathbb{D})$ tal que $f(\mathbb{D}) \subset \mathbb{H}$ y $f(0) = 1$ con desarrollo de Taylor $f(z) = 1 + \sum_{n=1}^{\infty} a_n z^n$. Entonces, $\forall n \in \mathbb{N} : |a_n| \leqslant 2$.*

***Demostración*:** Por la `prop-subordinacion-parte-real-positiva/`proposición 1, $\forall z \in \mathbb{D} : |f'(z)| \left(1 - |z|^2\right) \leqslant 2\Re(f(z))$. En particular, para $z = 0$, $|f'(0)| \leqslant 2\Re(f(0)) = 2$, luego $|a_1| = |f'(0)| \leqslant 2$.

Para $n \geqslant 2$, sea $\eta = e^{2\pi i/n}$ una raíz n-ésima de la unidad. Observamos que, $\forall k \in \mathbb{N}$:

$$\frac{1}{n} \sum_{j=0}^{n-1} \eta^{jk} = \begin{cases} 1, & \text{si } n \mid k, \\ 0, & \text{si } n \nmid k. \end{cases}$$

Por tanto, si consideramos la función $g\colon \mathbb{D} \to \mathbb{C}$ dada por

$$g(z) := \frac{1}{n} \sum_{j=0}^{n-1} f(\eta^j z) = \frac{1}{n} \sum_{j=0}^{n-1} \sum_{k=0}^{\infty} a_k \left(\eta^j z\right)^k = \sum_{k=0}^{\infty} a_k\, z^k \left(\frac{1}{n} \sum_{j=0}^{n-1} \eta^{jk} \right) = \sum_{\ell=0}^{\infty} a_{n\ell} z^{n\ell},$$

entonces $h(z) := \sum_{\ell=0}^{\infty} a_{n\ell} z^{\ell}$ define una función holomorfa en $\mathbb{D}$ tal que $g(z) = h(z^n)$. Además, $h(\mathbb{D}) \subset \mathbb{H}$ y $h(0) = a_0 = 1$. Por el caso $n = 1$, $|a_n| = |h'(0)| \leqslant 2$. ■

Observación 2. La cota del `cor-caratheodory-coeficientes-taylor-fn-holomorfa-parte-real-posit` es óptima, pues para $F \in \mathcal{H}(\mathbb{D})$ dada por $F(z) = \frac{1+z}{1-z}$, se tiene que $F(\mathbb{D}) \subset \mathbb{H}$, $F(0) = 1$ y $F(z) = 1 + 2 \sum_{n=1}^{\infty} z^n$.

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